deterministic Hamiltonian system taken first, followed by the Ornstein-Uhlenbeck equation for $\boldsymbol{p}$. The flows of each of these systems individually preserves the Gibbs distribution: denoting the Fokker-Planck operators associated to these flows by $\mathcal{L}_H^\dagger$ and $\mathcal{L}_{\mathrm{O}}^\dagger$ we have

$$\mathcal{L}_H^\dagger \rho_\beta = 0, \qquad \mathcal{L}_{\mathrm{O}}^\dagger \rho_\beta = 0,$$

and hence

$$\exp(t\mathcal{L}_H^\dagger)\rho_\beta = \exp(t\mathcal{L}_{\mathrm{O}}^\dagger)\rho_\beta = \rho_\beta.$$

We can consider the propagation of measure resulting from a numerical discretization defined by the composition of these two pieces. For a scheme $[\![\mathrm{HO}]\!]$, we have the distribution first propagated under Hamiltonian dynamics, and then the momenta are evolved as the solution to the Ornstein-Uhlenbeck process, which clearly will preserve the Gibbs measure

$$\exp(h\mathcal{L}_{\mathrm{O}}^\dagger)\left(\exp(h\mathcal{L}_H^\dagger)\rho_\beta\right) = \exp(h\mathcal{L}_{\mathrm{O}}^\dagger)\rho_\beta = \rho_\beta.$$

The problem with this "method" is of course that it requires the exact flow map of the Hamiltonian system. However, this requirement can be relaxed. Given any divergence free vector field $\boldsymbol{f}$, we have

$$-\nabla \cdot (\boldsymbol{f}\rho) = -\boldsymbol{f} \cdot \nabla \rho,$$

thus

$$\mathcal{L}_f^\dagger \rho = -\mathcal{L}_f \rho.$$

If $\mathcal{L}_f$ preserves a first integral I, then $\mathcal{L}_f$ preserves any distribution which is a function of I. It follows that any energy-preserving, divergence free vector field will always preserve the Gibbs distribution.

All symplectic methods preserve the volume, but for a system with no first integrals except the energy, it is known from a theorem of Ge and Marsden [400] that a symplectic method cannot preserve the system energy exactly (unless the symplectic method is itself a time-reparameterization of the exact solution). The options for volume preserving non-symplectic discretization methods that conserve energy exactly are limited (standard form schemes that rely on calculation of the vector field at a few points cannot achieve this [380]).

A class of methods that do provide the necessary features can be found in the work of Feng Kang [133], referred to as J-splitting by McLachlan and Quispel [261]. Let $\boldsymbol{J}$ be the skew-symmetric canonical symplectic structure matrix. The idea is to consider a splitting of $\boldsymbol{J}$ into a finite number K of skew-symmetric matrices $\boldsymbol{J}_i$, $i = 1, \ldots, K$. This induces a splitting of the Hamiltonian vector field into K vector